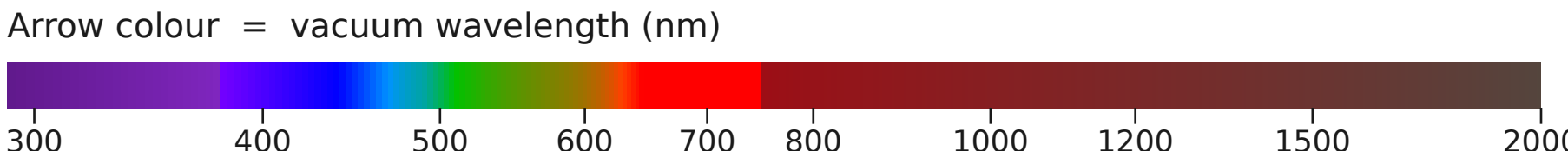


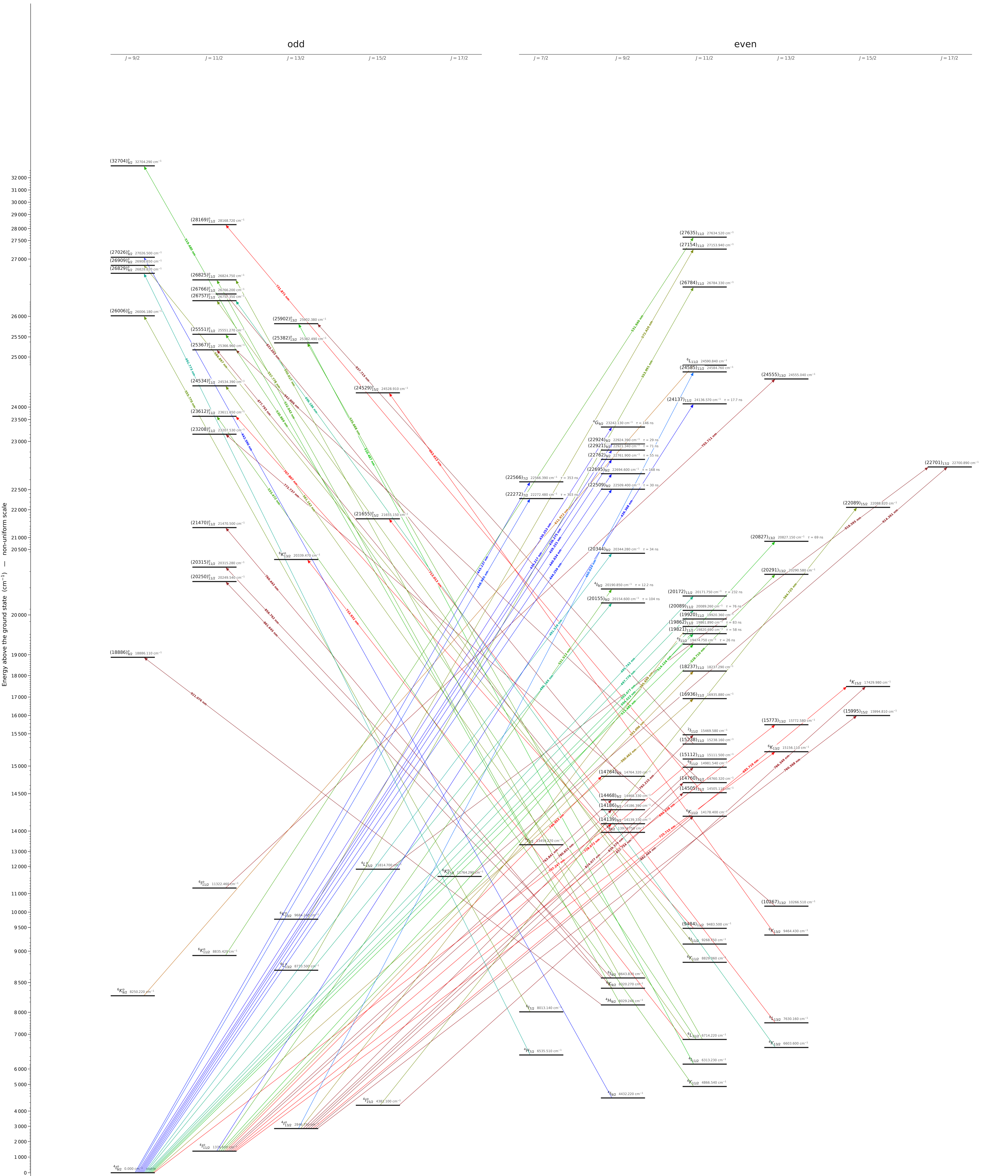
Praseodymium

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||e\mathbf{r}||J' \rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Pr (Pr I). Strongest classified transitions below 2 μm (NIST): 97 levels, 74 lines.

Pr⁺ ionisation limit 44120.00 cm⁻¹ = 5.47018 eV (226.655 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J e $\mathbf{r}$ J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁴ I _{9/2} - ⁴ G _{9/2}	430.2532	430.1322	696.7815	-	-	1.09	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (22924) _{9/2}	436.2166	436.0941	687.2559	-	-	5.488	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (22921) _{9/2}	436.2747	436.1521	687.1645	-	-	2.242	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (22762) _{9/2}	439.3306	439.2072	682.3846	-	-	2.894	-	excitation line used for the time-resolved laser-induced...
⁴ I _{11/2} - (24137) _{11/2}	439.3679	439.2445	682.3267	-	-	8.992	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (22695) _{9/2}	440.6335	440.5097	680.3670	-	-	0.9474	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (22566) _{9/2}	443.1369	443.0125	676.5234	-	-	0.4509	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (22509) _{9/2}	444.2588	444.1342	674.8148	-	-	5.305	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (22272) _{9/2}	448.9846	448.8586	667.7121	-	-	0.5253	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (20344) _{9/2}	491.5387	491.4015	609.9062	-	-	4.681	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (20172) _{11/2}	495.7428	495.6045	604.7338	-	-	0.686	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (20155) _{9/2}	496.1646	496.0262	604.2197	-	-	1.53	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (20089) _{11/2}	497.7784	497.6396	602.2609	-	-	2.094	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (19862) _{11/2}	503.4768	503.3364	595.4445	-	-	1.918	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - (19821) _{11/2}	504.5233	504.3827	594.2093	-	-	2.744	-	excitation line used for the time-resolved laser-induced...
⁴ I _{9/2} - ⁴ I _{11/2}	513.4854	513.3424	583.8983	-	-	6.121	-	excitation line used for the time-resolved laser-induced...
⁴ I _{11/2} - (20827) _{13/2}	514.1243	513.9811	583.1128	-	-	2.307	-	excitation line used for the time-resolved laser-induced...
⁶ I _{11/2} - ⁶ I _{9/2}	531.5120	531.3642	564.0370	-	-	13.05	-	excitation line used for the time-resolved laser-induced...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁴¹Pr hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (mHz)
⁴ I _{9/2}	926.209	-11.878	F=2: -12740.82 F=3: -9958.63 F=4: -6250.43 F=5: -1617.40 F=6: +3938.96 F=7: +10416.88
⁴ I _{11/2}	730.393	-11.877	F=3: -11873.80 F=4: -8948.60 F=5: -5293.56 F=6: -909.64 F=7: +4201.97 F=8: +10039.93
⁴ I _{13/2}	613.240	-12.850	F=4: -11503.19 F=5: -8434.04 F=6: -4750.48 F=7: -456.42 F=8: +4448.11 F=9: +9961.94
⁴ I _{15/2}	541.575	-14.558	F=5: -11513.77 F=6: -8259.83 F=7: -4465.46 F=8: -131.53 F=9: +4740.97 F=10: +10150.89
⁴ I _{9/2}	923.2(4)	-22(7)	F=2: -12704.08 F=3: -9927.88 F=4: -6228.85 F=5: -1609.18 F=6: +3928.37 F=7: +10380.50
⁴ K _{9/2}	867.997	-50.319	F=3: -14125.76 F=4: -10638.41 F=5: -6285.39 F=6: -1070.82 F=7: +3000.36 F=8: +11922.38
⁴ I _{11/2}	756.0(10)	F=3: -12285.00 F=4: -9261.00 F=5: -5481.00 F=6: -945.00 F=7: +4347.00 F=8: +10395.00	
⁴ H _{9/2}	979.0(10)	25(30)	F=1: -11000.36 F=2: -9049.50 F=3: -6120.54 F=4: -2210.25 F=5: +2685.64 F=6: +8572.50
⁴ K _{13/2}	755.456	-48.633	F=4: -14183.50 F=5: -10391.26 F=6: -5846.74 F=7: -553.31 F=8: +5485.10 F=9: +12264.00
⁴ L _{13/2}	474.692	-29.633	F=3: -7726.00 F=4: -5818.18 F=5: -3437.05 F=6: -585.01 F=7: +2735.00 F=8: +6519.61
⁴ L _{13/2}	776.286	-43.592	F=4: -14572.13 F=5: -10677.29 F=6: -6009.01 F=7: -570.31 F=8: +5635.28 F=9: +12603.75
⁴ I _{12/2}	168.0(10)	-	F=1: -1890.00 F=2: -1554.00 F=3: -1050.00 F=4: -378.00 F=5: +462.00 F=6: +1470.00
⁴ H _{9/2}	797.0(20)	-	F=2: -10958.75 F=3: -8567.75 F=4: -5379.75 F=5: -1394.75 F=6: +3387.25 F=7: +8966.25
⁴ K _{9/2}	213.531	-4.136	F=2: -2937.95 F=3: -2296.11 F=4: -1440.82 F=5: -372.47 F=6: +908.40 F=7: +2401.19
⁴ K _{9/2}	255.0(20)	-	F=2: -3506.25 F=3: -2741.25 F=4: -1721.25 F=5: -446.25 F=6: +1083.75 F=7: +2868.75
⁴ I _{9/2}	797.0(20)	-	F=2: -10958.75 F=3: -8567.75 F=4: -5379.75 F=5: -1394.75 F=6: +3387.25 F=7: +8966.25
⁴ I _{11/2}	854.297	-31.807	F=4: -16030.30 F=5: -11749.03 F=6: -6615.54 F=7: -632.04 F=8: +6198.91 F=9: +13874.37
⁴ K _{11/2}	769.0(10)	-30(20)	F=3: -12508.66 F=4: -9423.50 F=5: -5570.72 F=6: -952.79 F=7: +4427.34 F=8: +10566.25
⁴ K _{11/2}	949.091	-13.721	F=3: -15428.40 F=4: -11627.85 F=5: -6878.84 F=6: -1182.50 F=7: +5459.83 F=8: +13046.57
⁴ L _{13/2}	977.0(10)	-24(20)	F=3: -15886.18 F=4: -11970.85 F=5: -7079.63 F=6: -1214.49 F=7: +5622.22 F=8: +13427.75
⁴ K _{13/2}	1056.0(10)	-15(10)	F=4: -19805.77 F=5: -14521.15 F=6: -8181.52 F=7: -787.90 F=8: +7658.48 F=9: +17156.25
(9484) _{11/2}	731.0(10)	-15(10)	F=3: -11884.95 F=4: -8956.37 F=5: -5297.49 F=6: -909.52 F=7: +4206.05 F=8: +10047.50
⁴ K _{13/2}	991.907	-7.246	F=4: -18601.04 F=5: -13639.28 F=6: -7686.08 F=7: -741.95 F=8: +7192.52 F=9: +16116.68
(10267) _{13/2}	972.0(20)	-	F=4: -18225.00 F=5: -13365.00 F=6: -7533.00 F=7: -729.00 F=8: +7047.00 F=9: +15795.00
⁴ I _{11/2}	1272.0(10)	75(50)	F=3: -20638.98 F=4: -15573.89 F=5: -9233.32 F=6: -1611.14 F=7: +7300.02 F=8: +17508.75
⁴ L _{15/2}	355.0(20)	-	F=5: -7543.75 F=6: -5413.75 F=7: -2928.75 F=8: -88.75 F=9: +3106.25 F=10: +6656.25
⁴ I _{9/2}	854.0(10)	-15(15)	F=2: -11749.38 F=3: -9182.88 F=4: -5762.62 F=5: -1490.12 F=6: +3632.75 F=7: +9603.75
⁴ K _{11/2}	986.0(10)	-20(10)	F=3: -16030.77 F=4: -12060.66 F=5: -7145.48 F=6: -1226.86 F=7: +5673.23 F=8: +13552.50
(14186) _{9/2}	910.0(10)	-60(30)	F=2: -12540.00 F=3: -9792.00 F=4: -6135.00 F=5: -1575.00 F=6: +3880.50 F=7: +10222.50
(14468) _{9/2}	762.0(20)	20(20)	F=2: -10468.33 F=3: -8188.33 F=4: -5146.00 F=5: -1339.33 F=6: +3234.17 F=7: +8577.50
(14505) _{11/2}	777(3)	-	F=3: -12626.25 F=4: -9518.25 F=5: -5633.25 F=6: -971.25 F=7: +4467.75 F=8: +10683.75
(14764) _{9/2}	867.0(20)	-	F=2: -11921.25 F=3: -9320.25 F=4: -5852.25 F=5: -1517.25 F=6: +3684.75 F=7: +9753.75

Measured A, B constants; sources in the data file.